

## Project Documentation

### Project Executable Files

|               |                                        |
|---------------|----------------------------------------|
| Date          | 11 July 2026                           |
| Team ID       | SWTID-2026-9508                        |
| Project Name  | Credit Card Approval Prediction System |
| Maximum Marks | 2 Marks                                |

#### Purpose:

This document lists every executable and runnable file in the Credit Card Approval Prediction System repository, along with instructions to set up and run the project from scratch.

#### Executable Files:

| File                                   | Purpose                                                                                                             | How to Run                                       |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Flask/app.py                           | Main entry point of the web application. Loads the trained model and serves the prediction form and results.        | python app.py                                    |
| Notebooks/EDA_and_Model_Training.ipynb | Regenerates the trained model from raw data - EDA, preprocessing, training, evaluation, and saving model artifacts. | jupyter notebook (or Run All in Jupyter/VS Code) |
| requirements.txt                       | Lists all Python package dependencies required to run the project.                                                  | pip install -r requirements.txt                  |

#### Setup & Run Instructions:

- **Step 1:** Clone the repository and navigate into the project folder.
- **Step 2:** (Recommended) Create a virtual environment: `python -m venv venv`, then activate it.
- **Step 3:** Install dependencies: `pip install -r requirements.txt`
- **Step 4:** (Optional - model is already trained and saved) To regenerate the model, open and run all cells in Notebooks/EDA\_and\_Model\_Training.ipynb.
- **Step 5:** Navigate into the Flask folder: `cd Flask`
- **Step 6:** Run the application: `python app.py`
- **Step 7:** Open a browser and go to `http://127.0.0.1:5000`

#### Live Application - Prediction Form:

## INSTANT DECISION ENGINE

# Know your credit card decision in seconds, not days.

Enter the applicant's details below. Our model - trained and compared across four algorithms on historical approval data - returns an Approved or Rejected decision instantly, replacing slow manual review.



## Personal & Demographic Details

### Gender Category

Select

Anonymized category, as encoded in the training dataset.

### Age (years)

Select

### Marital Status Category

Select

### Ethnicity Category

Select

### Citizenship Category

Select

## Live Application - Prediction Result:

Example run with a strong applicant profile (high income, no prior default, employed):

[CC CreditApprove](#) Credit Card Approval Prediction System

✓

## Application Approved

Based on the details provided, the model predicts this application is likely to be approved.

Model confidence (probability of approval): 99.56%

[— Submit another application](#)

SmartBridge AI/ML Internship · Team SWTID-2026-9508 · Credit Card Approval Prediction System

*Screenshot captured from a live local run of the application, confirming the model returns a real Approved/Rejected prediction with a confidence score.*